

Food for Thought: Policy Approaches to Improving In-School Meal Participation

I. Problem Definition

In 2022, 12.8% (17 million) of U.S. households were considered by the U.S. Department of Agriculture (USDA) to be food insecure, meaning that they “were uncertain of having or unable to acquire enough food to meet the needs of all their members because they had insufficient money or other resources for food” (USDA, “Key Statistics & Graphics”). While these statistics alone are jarring, they are particularly worrisome when compared to those of previous years: in 2021 10.2% (13.5 million) (USDA, “Key Statistics & Graphics”) and in 2020 10.5% (13.8 million) of U.S. households were considered food insecure (Coleman-Jensen et al., 2020). Moreover, between 2008 and 2021 food insecurity was largely trending downwards; the recent departure from this trend underscores the pressing need to address food insecurity and prevent this issue from becoming worse (USDA, “Key Statistics & Graphics”).

One significant way that the government has sought to address food insecurity over the course of the past century is through meal programs in schools. In June 1946 Congress created the National School Lunch Program (NSLP), which established federal assistance for low-cost or no-cost lunches in public and nonprofit private schools (USDA, “National School Lunch Program Fact Sheet”). Through NSLP, children in families with income below 130% of the federal poverty level receive free school meals, and children in families with income between 130% and 185% of the federal poverty level are charged a reduced price for school meals (Ruffini, 2020); children in families with income above 185% of the federal poverty level can purchase a meal at the ‘full price,’ which is typically less than cost (Leos-Urbel et al., 2013). In its first year, approximately 7.1 million children participated in the NSLP; today, nearly 30 million children participate in the NSLP (USDA, “National School Lunch Program Fact Sheet”).

Moreover, in 1966 Congress passed The Child Nutrition Act, which piloted the School Breakfast Program (SBP) that eventually became permanent in 1975; the SBP has identical cutoffs to the NSLP for receiving free and reduced-price meals (Leos-Urbel et al., 2013). The SBP and the NSLP are both funded by federal grants that state education agencies may use to run these programs (Altindag et al., 2019). The NSLP and SBP are critical not only because of their capacity to address food insecurity, but also because they improve students' ability to learn. As expressed by Congressman John Flannagan, a co-sponsor of the National School Lunch Act, "If you want to dispel the gloom of Nazism and communism from the face of the earth, the thing to do is to feed and educate the peoples of those nations. A full stomach and a trained mind will never embrace either Nazism or communism" (Levine, 82). While the goals of education today look different than those referenced by Congressman Flannagan, it remains true that in order to achieve these goals, regardless of what they are, students must be properly fed.

While the NSLP and SBP have contributed significantly to addressing food insecurity among school-age children, these programs have several noteworthy faults that hinder their ability to fully address this issue. Firstly, while the NSLP is offered during a designated lunch period, the SBP is typically offered before school in the cafeteria, which requires students to arrive at school early in order to partake in the SBP - a requirement that may not be feasible for many students and families; this logistical challenge prevents some students from participating in the SBP, even though they would benefit from doing so. (Olarte et al., 2023). Another troublesome feature of these programs is that students who receive free or reduced-price meals make up a disproportionately large share of their participants. Recent statistics do not adequately illustrate this disproportionality because USDA pandemic waivers allowed meals to be provided free of charge to all students, regardless of their household's income; therefore, statistics from

before the pandemic paint a better picture of this imbalance (USDA, “School Breakfast Program”). For example, in 2019 approximately 73.1% of the NSLP participants received a free or reduced-price lunch and 84.1% of the SBP participants received a free or reduced-price breakfast (Toossi et al., 2021). However, less than 35% of children at this time were eligible for free or reduced-price meals (Semega et al., 2020). The disproportionate participation of children who qualify for free or reduced-price meals in either the NSLP or SBP is problematic because it paints these programs as programs for low-income students, which can discourage participation and prevent some children from eating breakfast or lunch. Students may also be reluctant to participate in these programs because in observing who receives free or reduced-price meals and who does not, they allow students to identify their peers’ socioeconomic statuses (Altindag et al., 2019). These deterrents are harmful not only because they prevent students from being adequately nourished, but also because consequently they prevent students from being able to meaningfully engage in the classroom. It is important that action is taken to encourage participation in the NSLP and SBP, to help ensure that school-age children are healthy and prepared to learn.

II. Policy Alternatives

While the NSLP and SBP are federally funded programs administered by the USDA, the actual funds for these programs are distributed by state education agencies, which has contributed to significant variation in how these programs are run. Consequently, though limited participation in the NSLP and SBP is a national issue, effective policy solutions to this problem have in some instances already been implemented locally. While typically, schools adhere to the means-tested meal programs, select schools have adopted alternative approaches to meal

provision that try to address issues of low participation and stigma. These local solutions can help inform federal policy that addresses low participation in the NSLP and SBP.

One productive policy alternative to the status quo - income-based pricing for both the NSLP and SBP - is the provision of universal free meals (UFM); this approach eliminates the income-based pricing of meals in schools and instead provides meals to all students at no cost (Ruffini, 2020). By eliminating the price stratification of school meals, UFM removes the financial burden and stigma associated with the NSLP and SBP; together, these effects help promote student participation in the NSLP and SBP.

One of the main channels through which schools have implemented UFM is through the Community Eligibility Provision (CEP) of the NSLP. Under this provision, “eligible” schools may receive federal assistance for reimbursement of UFM’s associated costs. Eligibility to participate in the CEP is determined by a school or district’s identified student percentage (ISP), the share of students who receive a form of income-based assistance, primarily Supplemental Nutrition Assistance Program (SNAP) and Temporary Assistance for Needy Families (TANF); schools and districts are considered eligible if their ISP reaches or exceeds an established minimum percentage (Ruffini, 2020). While initially the CEP only applied to select districts, this provision became nationally applicable in 2014 (Altindag et al., 2019). The CEP model illustrates one example of how UFM may be offered; however, decreasing or eliminating the ISP requirement may better promote student participation in the NSLP and SBP by expanding the network of schools that offer UFM and making school meal participation less challenging on a national level.

Beyond modifying the pricing scheme of the NSLP and SBP, other policy alternatives aim to change the way that the meals themselves are administered as a way of encouraging

participation in these programs. One such alternative policy is the Breakfast After the Bell (BATB) program. Under traditional SBP structures, students who are eligible for free and reduced-price meals make up a disproportionately large share of its participants - even greater than in the NSLP. The disproportionate share of low-income students among SBP participants is in part due to the logistical obstacles of getting to the cafeteria before the school day begins, an obstacle that is largely only worth surmounting for students who would otherwise be unable to afford breakfast. Through BATB, the SBP is offered to students free of charge and it is integrated into the school day; this reduces both the logistical and financial burdens of participating in the SBP, thus encouraging greater participation. BATB has been implemented through three principal models: Grab and Go Breakfast, Second Chance Breakfast, and Breakfast in the Classroom. Through Grab and Go Breakfast, prepackaged meals are distributed in high-traffic areas of the school to students as they arrive. Through Second Chance Breakfast, students can get breakfast from a designated location during an extended morning break. Through Breakfast in the Classroom, breakfasts are delivered directly to classrooms, often in insulated bags which maintain the temperature of food items and allow students to enjoy a nutritious breakfast (Olarde et al., 2023). While each particular BATB model offers unique advantages and disadvantages, they all demonstrate policy alternatives to how breakfast is served in schools that encourage student participation in the SBP and reduce the stigma associated with it.

III. Analysis of Policy Alternatives

Arguments in support of both UFM and BATB center around the important role that nutrition plays in children's ability to learn and succeed academically. As these programs both seek to improve the accessibility of school meals, advocates argue that they offer promising policy approaches to increasing student participation in the NSLP and SBP which can improve

students' health and academic success. While there is significant overlap in the effects and benefits of UFM and BATB, as they are separate policies I will discuss the arguments for and against each of them separately, beginning with UFM. One significant reason for advocates' support of UFM is its effectiveness in increasing school meal participation. In her study examining the effect that school-wide free meals have on uptake and student performance across many different districts and states, Krista Ruffini found that UFM increased breakfast and lunch participation by as much as 38% and 12% respectively (Ruffini, 2020). Similarly, in their study of the impacts that UFM had in New York City public schools, Amy Schwartz and Micah Rothbart found that not only did UFM increase school lunch participation, but this increase was greater among students who would otherwise not qualify for free or reduced-price lunch than among students who would (Schwartz and Rothbart, 2019). In a similar study focused on the impact that UFM breakfast had in New York City public schools, Jacob Leos-Urbel et al. found that UFM increased breakfast participation across all eligibility levels, but led to the greatest increase - 55% - in breakfast participation among students who would typically pay full-price (Leos-Urbel et al., 2013).

Advocates of UFM note that by significantly increasing school meal participation among students who would not usually qualify for free or reduced-price meals, this program helps address both the issues of uptake and stigma related to in-school meals. UFM makes school meals affordable for all students by making it easier for students who would have otherwise had to pay full-price for school meals to participate. Additionally, UFM eliminates the opportunity for school meal participation to publicize students' socioeconomic status. Both of these effects encourage participation in school meals among students of all socioeconomic backgrounds. The increased participation in school meals among full-price paying students in particular helps to

dispel the idea that these meals are for students from low-income households, and contributes to further increases in participation.

In addition to the significant increases in meal participation the UFM brings about, supporters of this policy highlight the impact that this program has beyond the cafeteria by improving students' health. By making it easier for students to participate in school meals, UFM helps ensure that students are properly nourished; as many scholars demonstrate, nutrition plays a key role in children's health. Improper nutrition may not only negatively impact children's physical development, but it can also increase their susceptibility to illness (Belot and James, 2011) and their probability of being obese (Schwartz and Rothbart, 2019). However, by increasing school meal participation, UFM helps to ensure that students are able to have nutritious meals that promote their physical wellbeing.

Another benefit associated with the increased meal participation from UFM that advocates highlight is the academic improvement that has been linked to increased school meal participation (Frisvold, 2014). In their study, Leos-Urbel et al. note that nutrition is critical for students' cognitive functioning: better nutrition has positive effects on recall, memory, concentration, and impulse control (Leos-Urbel et al., 2013). Therefore, by increasing school meal participation UFM not only helps ensure that students are adequately nourished, but it also helps promote the cognitive abilities of students so that they can be fully engaged in their classes. In her study of UFM in schools across the U.S., Ruffini found that UFM helped improve math performance, particularly for elementary school students (Ruffini, 2020). In their study of UFM in New York City public schools, Schwartz and Rothbart found that as a result of increased school lunch participation, students' performance improved in English Language Art and math (Schwartz and Rothbart, 2019). Furthermore, in their study on the impacts of the SBP, Judith

Bartfeld et al. found that while traditional SBP participation is associated with higher reading scores for boys, UFM breakfast increased the magnitude of such improvements and extended these benefits to a broader group of children (Bartfeld et al., 2019).

Not only has the increased school meal participation associated with UFM been shown to directly benefit students' academic performance, but scholars have also found that many other effects of increased meal participation indirectly improve students' academic performance. One prominent example is the impact that UFM has on behavioral incidents in school; Duha Altindag et al. examined the impact of UFM on behavior incidents and in turn academic achievement by studying the impact that a UFM program in South Korea had on misbehavior in schools.

Altindag et al. posited that because hunger can induce aggressiveness and impulsiveness, UFM programs, and their associated increases in school meal participation, may improve misbehavior in schools. Impressively, in their study Altindag et al. found that after implementing UFM programs, North Korean schools witnessed a nearly 35% reduction in misbehavior incidents, a large portion of which were physical fights between students (Altindag et al., 2019). Misbehavior and school violence has been demonstrated to reduce student academic performance and increase the propensity to drop out of school; therefore, by reducing misbehavior, especially incidents of school violence, UFM programs further promote students' ability to succeed academically (Altindag et al., 2019). Another indirect manner in which UFM helps improve students' academic performance is by promoting students' physical wellbeing. As previously discussed, inadequate nutrition can have adverse effects on children's health outcomes, which in turn contributes to a greater susceptibility to illness, resulting in more absences from school; missing school reduces the amount of instruction that students receive and decreases teacher contact, both of which hinder their academic performance (Belot and James, 2011). The advantageous health

outcomes that UFM promotes through increased meal participation can reduce days of absence and help ensure that students receive adequate instruction and teacher contact, which ultimately improve academic performance (Belot and James, 2011).

In ways similar to those of UFM programs, BATB advocates promote this policy because of its ability to increase SBP participation which consequently improves students' health outcomes and academic performance. By making in-school breakfast more accessible to students whether through Grab and Go Breakfast, Second Chance Breakfast, or Breakfast in the Classroom, BATB not only eliminates logistical barriers to participating in the SBP, but also helps decrease the stigma associated with the SBP. The rather significant burdens associated with the traditional SBP minimizes participation among students who do not rely on in-school breakfast for sustenance, which in many students' eyes brands the SBP as a program for students from low-income households. Thus, there is often a stigma associated with SBP that inadvertently discourages students from participating (Olarate et al., 2023). However, as a program that is easily accessible for all students, BATB is not perceived as a program for students of a particular socioeconomic background. In their study of BATB, Deborah Olarte et al. observed that invariably across states and districts, the SBP participation increased when breakfast was offered through one of the BATB models, in some schools contributing to an increase in participation as large as 38% (Olarate et al., 2023). These ubiquitous increases demonstrate the impact that BATB has on in-school breakfast participation by increasing SBP's accessibility and decreasing its stigma.

Much like UFM, increases in SBP participation that result from adopting BATB models have myriad beneficial effects; chief among them is improved academic performance, because with a substantive breakfast students are able to be focused (Olarate et al., 2023). In her study,

Ruffini notes that in-classroom breakfasts improve students' math and reading scores whereas traditional SBP serving methods do not (Ruffini, 2020). Similarly, Schwartz and Rothbart reference several studies that demonstrate greater math and reading score improvements associated with in-classroom breakfasts than with the traditional SBP model (Schwartz and Rothbart, 2019). Another positive effect of increased in-school breakfast participation from BATB is decreased student misbehavior. Across their examination of several studies analyzing the impacts of BATB, Olarte et al. consistently found that implementing this program led to improvements in student behavior which, as previously discussed, can in turn help improve academic achievement (Olarte et al., 2023). Lastly, by improving students' nutrition and increasing their access to food throughout the day, BATB has been shown to contribute to various health benefits such as improving students' immune systems and cognitive functioning (Olarte et al., 2023)

Arguments against both UFM and BATB revolve around similar themes: cost, administrative burden, and health concerns. Opponents of these programs often criticize them for their costliness, because they both eliminate revenue sources for in-school meal programs and require additional expenses to those of the traditional NSLP and SBP models such as appliances needed for BATB or increased provision of meals to meet increased demand under UFM (Altindag et al., 2019). Beyond the monetary costs of UFM and BATB, critics also cite the administrative costs imposed by these programs. In order to provide UFM, the USDA requires that schools keep track of the additional expenses incurred and then apply for reimbursement of its costs (Schwartz and Rothbart, 2019). This extensive bookkeeping would increase the demands on school administrations and in some cases it may exceed their administrative capacity. In addition to worrying about the costs of UFM and BATB, many critics also fear that

these programs will have detrimental effects on students' health. One such concern is that, as a result of BATB, students may eat breakfast at home and at school, which could lead to increased weight and BMI. Similarly, some critics fear that UFM may increase obesity by students either overeating or eating both a lunch they bring from home and a lunch provided to them in school (Olarate et al., 2023). Though UFM and BATB demonstrate significant beneficial effects, opponents find that both the cost and health concerns related to these programs outweigh their benefits.

IV. Recommendation

In light of the arguments laid out in section III, I would recommend a multi-phase policy solution that implements both UFM and BATB to address low participation and its adverse effects in both the SBP and NSLP. While UFM effectively addresses the issues of low participation and stigma associated with school lunch, it is limited in its ability to do the same for school breakfast. By eliminating both the logistical - in addition to the financial - burdens of participating in in-school breakfast, BATB has proven to be more effective than UFM alone in increasing student uptake of in-school breakfast. Therefore, I would recommend a policy that ultimately leads to the enactment of both UFM and BATB. Though both programs have demonstrated benefits for students' health and academic performance, UFM would likely have a broader impact as it improves the accessibility of both school lunch and breakfast. For this reason, I would recommend a phased approach to implement both of these programs that begins with the enactment of UFM. because it is important to improve students' nutrition overall, not just with respect to one meal, so that they can truly reap the benefits of adequate nourishment

For the first phase of my recommendation, I propose introducing UFM in public schools to kickstart the growth of participation rates in both the NSLP and SBP. As the successful

implementation of any new program in schools takes time, often requiring schools to adjust to new practices and address issues that arise as a result of those new practices, this first phase should last for as long as schools need to grow accustomed to the provision of UFM. Once a school demonstrates its ability to successfully administer UFM, phase two of implementing BATB should begin. At the beginning of this phase, schools should identify which BATB model they will follow, after which schools should acquire the equipment needed to adopt their model of choice and thereafter serve BATB.

As many of the scholars previously cited have demonstrated, students' lack of participation in in-school meals may stem from a lack of financial resources, but it may also be a result of the stigma associated with such meals. By improving the accessibility of both in-school breakfast and lunch, this phased approach helps to address both the stigma and financial burden associated with participation in both the NSLP and SBP; if successfully implemented, by increasing participation in the NSLP and SBP, this approach should subsequently bring about the myriad benefits associated with improved nutrition. However, as critics note, this approach would be costly; under the current structure that the USDA uses for the implementation of UFM through CEP, the increased expense of school meals in the first phase would likely fall entirely on the federal government. While this expense is certainly significant, the cost effectiveness of UFM and BATB as a means of improving academic achievement, in addition to their benefits for students' health, justifies the seemingly high price. In their study, Schwartz and Rothbart found that UFM was an extraordinarily cost effective way for New York City public schools to improve academic achievement; they found that with an additional \$50 spent per student for UFM, schools saw math and English Language Arts test scores improve by 0.1 standard deviations (Schwartz and Rothbart, 2019). Moreover, Ruffini observed that across the many

different states and districts that she studied UFM offered benefits similar in magnitude to other forms of federal government assistance at a fraction of the cost: while the UFM programs she examined cost the federal government approximately \$100 per student, they delivered the same magnitude of test score improvement that results from a \$500 Earned Income Tax Credit (Ruffini, 2020). A similar line of reasoning may also be used for BATB, which would require either federal or state funding for the new appliances needed in order to carry out this program. Though the program itself would be expensive to implement, it has been shown that providing breakfast in the classroom rather than in the cafeteria increases both math and reading test scores (Schwartz and Rothbatt, 2019). Therefore, despite being a high cost meal program, BATB can also be understood as a cost efficient manner of improving academic achievement. While it is certainly true that UFM and BATB would increase government spending on the NSLP and SBP, the benefits of these programs would likely lead to savings in other areas of government spending, ultimately justifying the cost of these programs.

Once implemented, this policy should significantly expand students' access to food; while in many cases this would be beneficial, ensuring that students are adequately nourished, critics fear that this increased accessibility of food will lead to increases in students' weight and BMI. However, evidence suggests not only that this claim lacks data to support it, but moreover that data suggests that the opposite may actually be true. In studying the impacts that UFM had in New York City public schools, Schwartz and Rothbart found that as student participation in the NSLP increased, their probability of being obese as well as their BMI decreased (Schwartz and Rothbart, 2019). While this correlation has not been uniformly substantiated in other studies, collectively the body of research that examines the impact of UFM and BATB on BMI and weight status indicates that there is mixed evidence about whether these programs lead to

beneficial changes in students' weight or BMI (Olarate et al., 2023). Nonetheless, evidence does not suggest that UFM or BATB have detrimental effects on students' weight or BMI, indicating that fears about these programs causing weight gain are unfounded.

V. Implementation Issues

Due to the concerns about cost and weight gain that may be associated with UFM and BATB, it is likely that my policy recommendation will face opposition among parents and select congressional or local government representatives who are opposed to increasing government funding for the NSLP or SBP. The fears among parents and others who are worried about the impact of these programs on children's weight can be addressed effectively by presenting and publicizing not only the data that disproves these concerns but also data which demonstrates that these programs actually have beneficial health impacts such as improved cognitive functioning and decreased susceptibility to illness. Similarly, the fears among congressional and local government representatives as well as members of the public who are against increasing government funding for school meals could be effectively addressed by communicating their cost effectiveness as a means of improving students' academic performance; cost benefit analyses could help further illustrate the value of investing in UFM and BATB.

Additionally, schools may be concerned about their administrative capacity to manage the applications for reimbursement of the costs of these programs or their ability to meet increased demand for in-school meals. While these concerns are legitimate, it seems that the benefits both students and schools would derive from UFM and BATB are well worth the hurdles that they may encounter from implementing these programs. Though it may take several years for schools to adjust to the changes that accompany UFM and BATB, once they have

grown accustomed to offering these programs they are likely to see improvements in their students' health and academic performance.

VI. Evaluation

The implementation of UFM and BATB will each lead to distinct short and long term changes. If their implementation follows my two phase recommendation, for the first two or three years only UFM will have been put in place; during this period schools should start to see an increase in breakfast and lunch participation, with the former likely being smaller than the latter. During this time school attendance should also increase and the frequency of behavioral incidents should decrease, as both of these issues have been demonstrated to improve alongside in-school meal participation. Once BATB has been implemented, schools should continue to see improvements in all of these areas, particularly in breakfast participation. In the long term, schools should see academic performance improve, however this change will likely occur gradually and manifest itself more prominently after both UFM and BATB have been implemented. While aimed at increasing in-school meal participation, if successfully implemented, both UFM and BATB will lead to many other positive impacts on students' health and academic performance which can help address several other issues within the education system.

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